PATHOLOGY STUDY EDITION

CHAPTER 08

ROBBINS & COTRAN • MECHANISM-FIRST REVIEW

Infectious Diseases

From portal of entry to tissue pattern: a visual, high-yield guide to how microbes persist, injure, and evade immunity.

Pathogenesis

ENTRY • SPREAD • DAMAGE

Core prototype

TUBERCULOSIS

Clinical anchors

VIRUSES • FUNGI • PARASITES

The infection playbook

Disease is the outcome of a three-way interaction: microbial virulence, route of exposure, and the strength or failure of host defenses.

1. ENTRY
skin • lung • gut • GU

→

2. ESTABLISH
adhere • invade • evade

→

3. SPREAD
local • lymph • blood

→

4. DAMAGE / EXIT
toxins • immunity • transmission

Portal barriers

- **Skin:** keratin, low pH, fatty acids, defensins. Trauma, lines, bites and vectors breach it.
- **GI:** gastric acid, mucus, bile, enzymes, microbiota. Acid neutralization markedly increases susceptibility.
- **Respiratory:** mucociliary clearance, cough, macrophages. Small aerosols reach alveoli.
- **GU:** urine flow and mucosal defenses; obstruction and instrumentation increase risk.

Microbial virulence moves

- Adhesins and biofilms permit attachment and device persistence.
- Capsules, antigenic variation, and intracellular niches evade elimination.
- **M. tuberculosis** survives by blocking phagolysosome formation.
- Exotoxins have focused effects; endotoxin drives cytokines, shock, and DIC.

Host response: defense with a cost

Neutrophils control pyogenic bacteria but create pus and collateral tissue injury. Th1/macrophage responses contain intracellular organisms yet build granulomas and fibrosis. Cytotoxic lymphocytes remove virus-infected cells but can injure the infected organ. Immunodeficiency shifts the pattern toward severe, disseminated, opportunistic disease.

Exam anchor: A true pathogen is not normal microbiota. Commensals become pathogenic when barriers fail or immunity is impaired.

Read the tissue reaction

The dominant inflammatory pattern is a practical shortcut to the class of infectious agent and the immune response involved.

PATTERN	MECHANISM	HIGH-YIELD CLUE
Suppurative	Neutrophil-rich inflammation with necrotic debris.	Pyogenic bacteria; abscesses and pus.
Granulomatous	T cells activate macrophages against persistent intracellular agents.	TB, fungi, parasites; epithelioid histiocytes/giant cells.
Cytopathic	Viral replication causes death, fusion, inclusions, or proliferation.	HSV inclusions, CMV “owl eye,” HPV koilocytosis.
Necrotizing	Toxins, vascular injury, or overwhelming inflammation.	Clostridia, severe viruses, angioinvasive fungi.
Chronic / scar	Persistent antigen couples inflammation to repair and fibrosis.	TB, HBV/HCV, schistosomiasis.

Viral injury

Direct lysis, apoptosis, altered cell function, transformation, and CTL-mediated injury. Persistence may be **latent**, **chronic productive**, or **transforming**.

Bacterial injury

Adherence and invasion, intracellular survival, exotoxins, endotoxin-driven inflammation, and immune-mediated tissue damage.

Persistence, latency & signature lesions

For every virus, learn the tissue tropism, the host at risk, and the diagnostic morphologic signal.

Acute viral syndromes

Measles: respiratory spread, cough/coryza/conjunctivitis and rash; giant cells/inclusions. Severe in impaired cellular immunity.

Dengue: febrile syndrome with vascular permeability; severe disease produces hemorrhagic shock.

SARS-CoV-2: severe disease shows diffuse alveolar damage and inflammation.

Herpesvirus latency

HSV: painful vesicles/ulcers; Cowdry type A intranuclear inclusions; latency in sensory ganglia.

VZV: varicella followed by latency; reactivation gives dermatomal zoster.

CMV: opportunistic disease in transplant or advanced HIV; “owl-eye” nuclear inclusions.

EBV and B cells

EBV infects B cells through **CD21**. In mononucleosis, atypical CD8+ T cells control proliferation. Weak T-cell immunity permits uncontrolled B-cell proliferation and lymphoma risk.

Three persistence modes

Latent: genome persists with minimal production (herpesviruses).
Chronic productive: ongoing replication (HBV/HCV).
Transforming: altered growth control (HPV, EBV-associated disease).

Pattern recognition: Immunodeficient host + enlarged cells with nuclear inclusions = think CMV. Vesicles + Cowdry A = HSV/VZV.

Containment creates the lesion

TB is the chapter’s prototype intracellular infection: the immune response restrains the bacillus, yet also causes caseation, fibrosis, and cavitation.

INHALE
alveolar macrophage entry

→

EARLY PHASE
intracellular replication & seeding

→

~3 WEEKS
Th1 response, IFN-gamma

→

OUTCOME
granuloma, control or tissue injury

Primary TB

First exposure in an unsensitized host. Usually contained; a Ghon focus with hilar node involvement may heal with fibrosis and calcification. Progressive disease occurs with weak host defense.

Secondary TB

Reactivation or reinfection in a sensitized host. Classically apical; the brisk tissue response causes prominent necrosis and **cavitation**, facilitating transmission.

Immune axis

Th1 cells activate macrophages. **IFN-gamma** makes them microbicidal; **TNF** maintains granulomas. AIDS, TNF blockade, diabetes and malnutrition favor active disease/reactivation.

Testing logic

Active disease: smear/culture plus nucleic-acid testing. Latent infection: **IGRA / PPD** detects sensitization only - it does not prove active disease.

Memory line: IFN-gamma turns the macrophage on; TNF holds the granuloma together.

Link organism to host defect

Instead of memorizing disconnected lists, start with the failed arm of immunity and predict the pathogen.

Fungal danger signals

Candida: mucocutaneous disease after antibiotics or immune dysfunction; invasive disease with neutropenia/devices.

Aspergillus: septate acute-angle hyphae; angioinvasion causes infarction and hemorrhage.

Mucor: broad nonseptate right-angle hyphae; angioinvasive in ketoacidosis/neutropenia.

Parasite anchors

Malaria: liver then RBC infection; *P. falciparum* causes severe disease through sequestration of infected RBCs.

Toxoplasma: congenital infection and encephalitis in advanced HIV.

Schistosoma: eggs drive granulomas and fibrosis; larvae penetrate intact skin.

The five takeaways to own

- Portal defenses explain exposure risk and which microbes gain entry.
- Purulence means neutrophil-driven pyogenic inflammation; granulomas signal persistent intracellular pathogens.
- TB pathology reflects Th1-mediated containment and collateral injury.
- Angioinvasive fungi create ischemic, necrotic, and hemorrhagic lesions.
- Host immune defect is the fastest path to predicting opportunistic infection.

Study prompt: For each infection, ask: portal of entry → immune evasion → tissue pattern → host at risk → diagnostic clue.